JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 105, NO. B10, PAGES 23,595–23,603, OCTOBER 10, 2000

Paleomagnetism of Siluro-Devonian sequences, NE Spain

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Abstract. The paleogeography of northeastern Spain is addressed in a detailed paleomagnetic study of Siluro-Devonian boundary sediments from the Catalan Coastal Ranges (CCR) at Santa Creu d'Olorda, Barcelona, and the eastern Pyrenees at Baèn. The sequences studied are all well dated biostratigraphically and have undergone only very minor deformation, mainly during the late Carboniferous, and extremely low grades of metamorphism (anchizone or less). After stepwise thermal demagnetisation of samples and principal component analysis of the data, up to two paleomagnetic directions can be clearly identified after removal of a present-day magnetic overprint. The first component (labeled B) is identified only in samples from the CCR with intermediate unblocking temperature spectra (300°-400°C). An overall mean direction of 187°/-03°, semi angle of cone of 95% confidence α_{95}=9.0°, and precision parameter k= 43.1 in situ, and 195°/03° α_{95}=24.4°, and k= 7.1 (51 samples at seven sites) after bedding correction is obtained for B. The exact age of this secondary component is uncertain, but when compared to published paleomagnetic data, a late Carboniferous to Early Triassic age could be inferred. A stable endpoint direction (labeled C) can be identified between 400° and 600°C in samples from both regions. For the CCR it is southwesterly directed with intermediate inclination values in situ, and yields a mean direction of 208°/48°, α_{95}=12.8° and k=28.4 (six sites) after bedding correction. It passes the fold test, is considered to be primary in origin, and is thus indicative of the Siluro-Devonian paleofield direction in the CCR. In the Pyrenean samples the C direction is northwesterly directed in situ, with a mean direction of 240°/51°, α_{95}=6.5° and k=86.5 (seven sites) after bedding correction. Although there is a clear improvement in grouping of these site mean directions, this is not significant at the 95% level of confidence. Combining the site mean directions obtained for component C from both regions yields an overall mean direction of 224°/51°, α_{95}=8.5° and k=24.6 (13 sites) after bedding correction. The improvement in grouping is significant at the 99% level of confidence at 90% unfolding, reinforcing the interpretation that this is a primary magnetisation. Adopting the reversed polarity option, the inclination value obtained translates into paleolatitudes of 30°S, placing NE Spain in the same latitudinal belt as the Armorican Massif and the Barrandian terrane of the Bohemian Massif (i.e., the Armorican Terrane Assemblage) in latest Silurian/earliest Devonian times.

1. Introduction

Pre-Mesozoic geology of the Iberian Peninsula is very complex, and a number of different tectonostratigraphic zones have been defined (Figure 1), namely, the Cantabrian Zone, West-Asturian-Leonese Zone, Central Iberian Zone, and the Ossa Morena and South Portuguese Zones. The South Portuguese Zone is clearly an exotic terrane which accreted onto Iberia in early Carboniferous times [*Silva et al.*, 1990; *Quesada et al.*, 1991]. Tectonic affinities of the Ossa Morena Zone, southern Spain, are more equivocal. On the basis of faunal differences it has been proposed that the Badajoz-Cordoba shear zone is a Variscan suture zone separating the Ossa Morena Zone from central Iberia [*Robardet*, 1976; *Robardet and Gutierrez-Marco*, 1990]. Alternatively, the Ossa Morena Zone may simply be the southern (in present-day coordinates) imbricated margin of the Iberian terrane. With the exception of allochthonous nappes of unknown affinity in the northwestern part of the Central Iberian Zone, the other zones of the Iberian Peninsula are considered to have formed a coherent tectonic block within the Variscan system, the Iberian Terrane [after *Quesada et al.*, 1991]. In general, the lithological characteristics of the Iberian Terrane, in terms of paleogeography, indicate high paleolatitudes in the Ordovician with clastic sedimentation and Late Ordovician glacial deposits, and significantly lower

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Paper number 2000JB900111.
0148-0227/00/2000JB900111$09.00

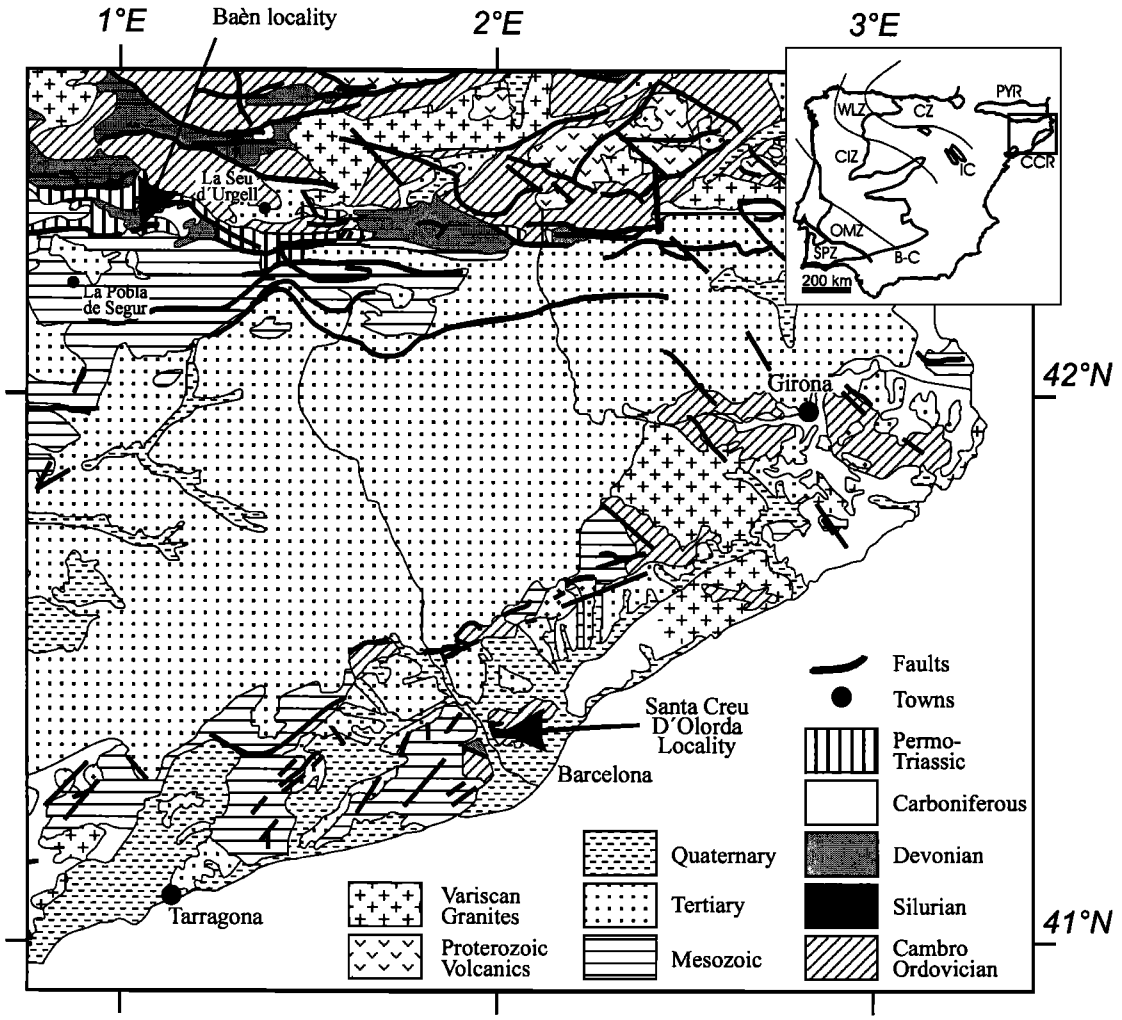


Figure 1. Geological sketch map of northeastern Spain. Inset shows the subdivision of the Iberian Variscan belt after *Julivert et al.*, [1974]. CZ, Cantabrian Zone; WLZ, West-Austrian-Leonese Zone; CIZ,Central Iberian Zone; OMZ, Ossa-Morena Zone; SPZ, South Portuguese Zone; PYR, Pyrenees; CCR, Catalonian Coastal Ranges; IC, Iberian Chains (Celtiberia); B-C, Badajoz-Cordoba shear zone.

paleolatitudes by the Devonian with the development of carbonate reef deposits. While strong similarities between Ordovician to Devonian successions in the Armorican Massif and central Iberia demonstrate that these two regions were closely related during the early and middle Paleozoic, details of their drift history and the tectonic relationship with other Paleozoic massifs and blocks within the Variscan fold belt remain poorly constrained.

To date, most paleomagnetic studies of the Iberian Terrane have concentrated on the Cantabrian Arc of northern Spain. Although no primary magnetisations have yet been recognized in these sequences, both geological and paleomagnetic evidence suggest that oroclinal bending of this arc occurred during the closing stages of the Variscan orogeny, possibly even into the Triassic [*Franke*, 1989; *Matte*, 1991; *Quesada et al.*, 1991; *Hirt et al.*, 1992; *Ries and Shackleton*, 1976; *Ziegler*, 1990; *Parés et al.*, 1996]. Paleomagnetic results available from elsewhere in the Iberian Massif are from Early Ordovician age sequences from northern Spain (Cabo de Peñas [*Perroud*, 1983]) and central Portugal (Buçaco red beds [*Perroud and Bonhommet*, 1981]). Although neither of these results are of particularly high quality (both have poor age constrains, and the demagnetisation and analytical techniques are not up to modern standards), they indicate high southerly paleolatitudes, as would be expected from the geological record. *Perroud et al.* [1991] published paleomagnetic results from Siluro-Devonian age sequences from the Almaden in central Spain. These data, however, have subsequently been shown to represent remagnetisation directions [*Parés and Van der Voo*, 1992]. The only other results available are those of *Perroud and Bonhommet* [1984] from Late Silurian age rocks of the San Pedro region. These data indicate fairly shallow paleolatitudes (~20°S); the age of magnetisation, however, is only rather loosely controlled and, as stated by the original authors, could have been acquired any time between Late Silurian and mid-Carboniferous times.

Detailed stratigraphic correlation between many of the Paleozoic terranes now incorporated within the Alpine fold belt and along the southern margin of Europe is particularly difficult due to the often high degree of deformation and metamorphism. Paleozoic sequences in the eastern Pyrenees and Catalan Coastal Ranges (CCR) are further complicated

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due to the lack of diagnostic fossils in the pre-Caradocian rocks. Nevertheless, geological studies demonstrate that there are many faunal similarities between the various Paleozoic terranes now located along the southern margin of Europe, i.e., the Montagne Noire, Massif de Mouthoumet, the Northern Greywacke Zone (northern Alps), and the Carnic Alps. Facies studies suggest that from Ordovician until at least mid-Devonian times, deposition in these regions took place in a passive margin environment, with a stable hinterland to the south and a deepening basin located to the north [*Echtler*, 1990; *Feist et al.*, 1994; *Julivert et al.*, 1987; *Matte*, 1991; *Heinisch et al.*, 1987; *Quesada et al.*, 1991]. Furthermore, the Devonian pelagic nodular limestone facies (Griotte facies) which are typical for the Pyrenees and the Montagne Noire are in marked contrast to the general shallow water shelly carbonates found in the Iberian and Armorican Massifs [*Julivert et al.*, 1987]. These differences are strong enough to suggest a separate facies province. Faunal assemblages from the Cantabrian region, Catalunya, and the Pyrenees are also appreciably different from other regions of the Ibero-Armorican Arc; thus the Paleozoic sequences in northeastern Spain and southern France may have been independent of the Iberian terrane, possibly forming part of an indentor block which caused bending of the Ibero-Armorican Arc in late Variscan times [*Riding*, 1974; *Robardet and Guiterrez-Marco*, 1990]. Paleogeographic affinities, however, are difficult to deduce from the geological record alone, and it remains unclear if the Catalan Coastal Ranges and the Axial Zone of the Pyrenees (here collectively termed the Catalan Terrane) were part of the Iberian Massif in the early Paleozoic, if they formed an independent microplate, if they formed part of the Armorican Terrane Assemblage, or if they remained adjacent to the northern margin of Gondwana. In addressing these questions, a detailed paleomagnetic study of Siluro-Devonian boundary rocks from the Catalan terrane has been carried out.

2. Local Geology and Paleomagnetic Sampling

The oldest exposed sediments of the CCR and the eastern Pyrenees, NE Spain, are generally classed together as Cambro-Ordovician in age (Figure 1), and comprise a monotonous sequence of greywackes, sandstones, and phyllites with minor limestone horizons. The Silurian is represented by sequences of black graptolite shales, ubiquitous throughout Europe, which often acted as major décollement horizons due to ductility contrast during Variscan deformation in the Pyrenees. These sediments are in turn overlain by a continuous sequence of Devonian through to early Carboniferous limestones and calcareous shales. The main phase of deformation in both regions was in the late Carboniferous, and the folded Paleozoic sequences are unconformably overlain by Westphalian to Stephanian age volcanic and sedimentary rocks. In general, the Variscan structures in these regions show little evidence for pervasive Alpine deformation.

Although Paleozoic rocks are fairly extensively exposed in the CCR (Figure 1), they tend to be rather highly deformed and metamorphosed such that Cambrian to Lower Silurian rocks are too highly metamorphosed and deformed to be suitable for paleomagnetic research. Upper Silurian to mid-Carboniferous sediments, although poorly exposed, have undergone, at most, anchizone metamorphism and were only very gently folded during the mid Upper Carboniferous (Westphalian) deformation. Upper Silurian and Lower Devonian sequences are exposed in a working quarry at Santa Creu d'Olorda, near Barcelona, where two formations can be recognized. The lower formation, dated as Upper Ludlovian to Lower Lochkovian (conodont stratigraphy), comprises massive nodular limestones, whilst the upper formation (Lochkovian to Emsian graptolite zones) consists of alternating marls and limestones with pelagic and hemipelagic faunas [*Garcia-López et al.*, 1990]. Samples were collected from seven sites in the Santa Creu d'Olorda quarry, covering a stratigraphic interval from the lower Pridolian carbonate sequences through to the Lockhovian marly limestones with a stratigraphic thickness of some 200 m.

In the eastern Pyrenees, Paleozoic sediments and basement rocks, which were uplifted during the Tertiary, are now exposed extensively in the Axial Zone. Although one of the most pronounced characteristics of these sequences are their high-temperature/low-pressure metamorphism, in the southern border of the Axial Zone metamorphic grades are much lower, in places reaching, at most, only anchimetamorphism [*Zwart*, 1986]. Furthermore, the higher stratigraphic levels in general have undergone much less deformation and lower grades of metamorphism, and the Devonian to Carboniferous sediments have not undergone any significant tectonic burial. The Silurian is again represented by graptolite shales, overlain by a complete sequence of Devonian through to early Carboniferous carbonates. There is no evidence for any deformation event prior to the late Carboniferous, and Westphalian D to Stephanian age sediments unconformably overlie the older folded sequences. In general, Tertiary deformation and metamorphic overprinting of the basement is only very weak and confined to major fault zones [*Wickham and Oxburgh*, 1986]. Devonian age sediments are very well preserved in the Axial Zone, and a number of faunal studies, in particular conodont biostratigraphy, provide rather precise relative age constraints. Earliest Devonian outcrops are extremely limited, and sampling was concentrated near the village of Baén, southwest of La Seu d'Urgell (Figure 1). These rocks comprise dark grey, medium bedded limestones and shales of the Gedinnian (Lochkovian) age Rueda Formation (Sierra Negra subfacies). Samples were collected from nine sites (72 samples) over a stratigraphic thickness of some 150 m and were subjected to detailed thermal demagnetisation experiments.

All samples were collected using a portable petrol-driven drill and oriented in the field using Sun and/or magnetic compasses. Samples were then stepwise thermally demagnetised, and magnetisations were measured using a 2G cryogenic magnetometer. Results were analyzed using principal component analysis with orthogonal [*Zijderveld*, 1967] and stereographic projection of the data. *Fisher* [1953] statistics were used in all cases. Although none of the samples have undergone any apparent penetrative deformation, the anisotropy of magnetic susceptibility was measured on a number of samples using a KLY-2 Kappabridge. The results show that only very low degrees of magnetic alignment (~1%) are present, indicating that despite their complex

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structural history, no secondary fabrics have been impressed on the rocks.

3. Paleomagnetic Results

During thermal demagnetisation experiments of samples from the eastern Pyrenees, two directions of magnetisation can be clearly identified in samples from seven out of nine sites. The first has fairly low unblocking temperatures (generally removed below 300°C) and is termed the A magnetisation direction. In most cases this component is northward directed with fairly steep positive inclination values in situ and is interpreted to reflect a present-day overprint direction. In a very small number of samples a northeasterly component of magnetisation can be identified, again with fairly steep inclination values. This may reflect an overprint direction related to deformation during the Alpine orogeny, but it is not possible to clearly isolate this from the present-day overprint direction. Both directions are carried by minerals with similar magnetic properties, and the extent to which the present-day component overlaps the northeasterly direction cannot be determined. Thus, no geodynamic interpretation of the north easterly direction can be made in terms of tectonic rotations related to younger deformation events.

The second, higher-temperature, direction of magnetisation identified in the Pyrenean samples, termed the C magnetisation, is generally very well defined in the unblocking temperature range of 300°-580°C and is directed towards the origin in orthogonal projections of results (Figure 2). Intensity decay diagrams generally show a gradual decrease in magnetic intensity from room temperature up to 580°C by which time all but two of the samples measured are demagnetised to <5% of their original intensity. A number of samples also show a slight inflection in the intensity decay curves between 300° and 350°C. These results suggest the predominance of magnetite as the principal magnetic carrier with minor amounts of pyrrhotite contributing to the natural remanent magnetization (NRM). The higher-temperature component of magnetisation (C direction) is identified in seven sites (BAE2-BAE4 and BAE6-BAE9) with single polarity, and is northwesterly directed with fairly steep positive inclination values in situ (Figure 3). Of the remaining two sites (BAE1 and BAE5), the directions were either extremely scattered within site (BAE1) or were unstable or extremely poorly defined (BAE5) due, most likely, to effects of weathering as was noted at the outcrop.

Samples from the CCR show fairly uniform behavior. Natural remanent magnetisation intensities are fairly low (generally less than 0.8 mA/m), and up to three directions of magnetisation can be identified. The lowest-temperature component (termed component A, generally removed below 200°C) is close to the direction of the local present-day Earth's magnetic field and is most likely a recent viscous overprint. An intermediate unblocking temperature component (200°-450°C), component B, is identified with single polarity in most samples from all seven sites from the CCR. It is southerly directed with very shallow inclination values in situ (Figures 4 and 5, and Table 1).

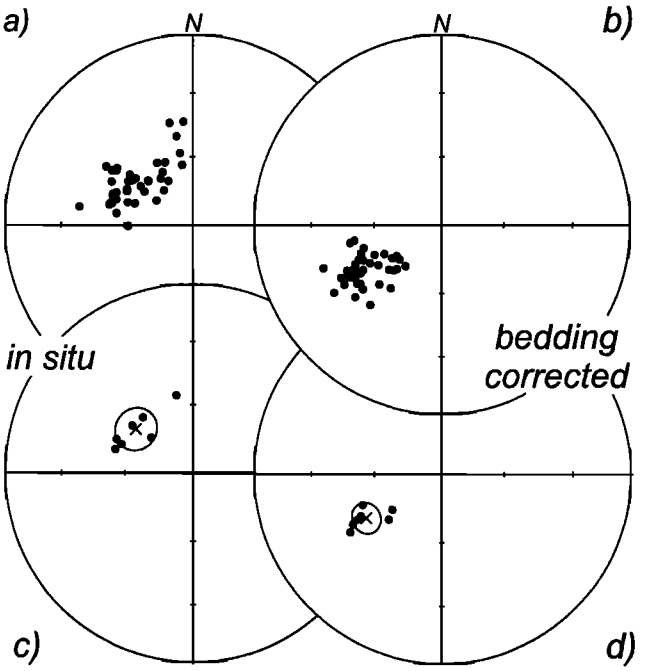


Figure 3. Equal area projection of component B from the Catalan Coastal Ranges. Solid (open) dots indicate lower (upper) hemisphere projection.

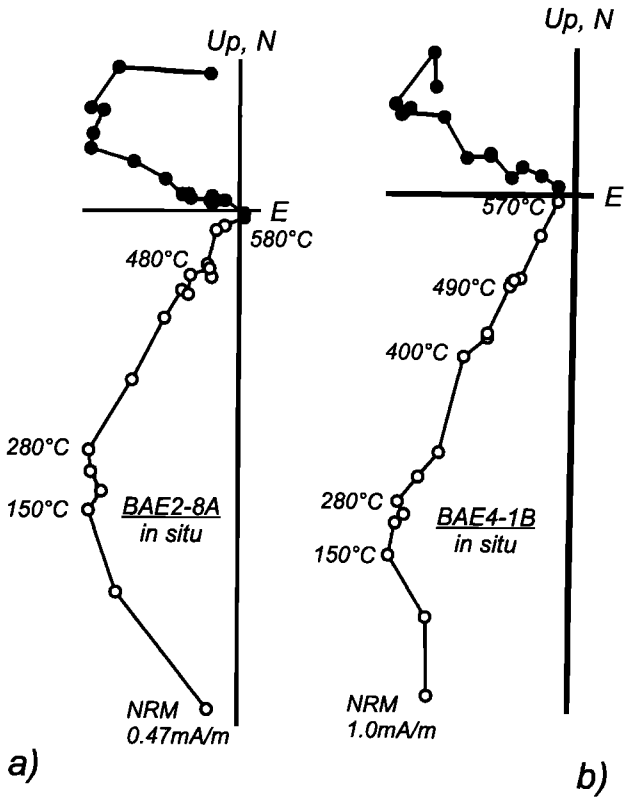


Figure 2. Orthogonal projection of thermal demagnetization characteristics from the Catalan Coastal Ranges. All directions are plotted in geographical coordinates, and solid (open) circles represent the horizontal (vertical) component. These examples show removal of component B (200°-500°C) followed by removal of component C at higher temperatures as stable endpoint directions.

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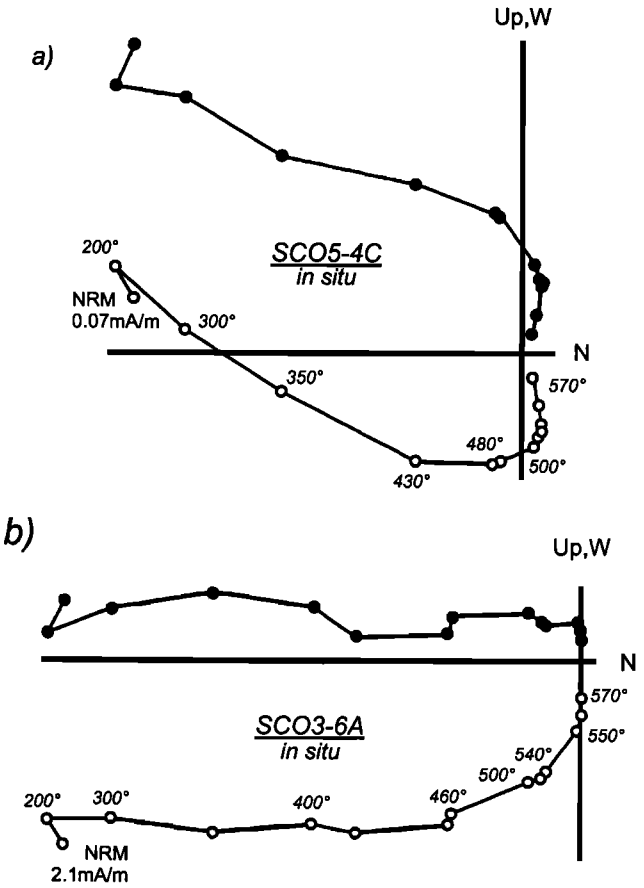


Figure 4. Equal area projection of the sample directions and site mean directions, both before and after bedding correction, obtained from the Catalan Coastal Ranges. Notation is the same as for Figure 3.

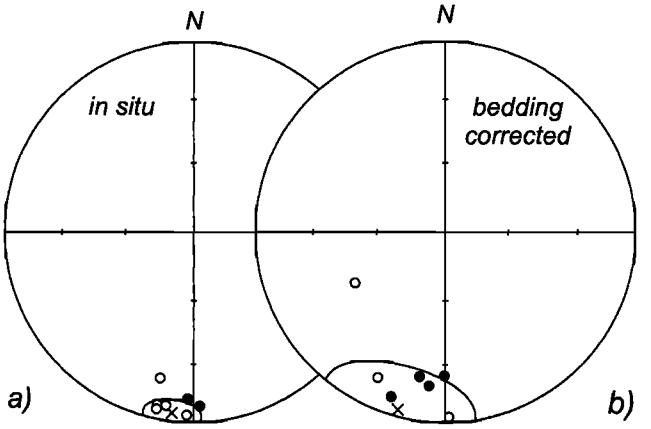


Figure 5. Thermal demagnetization characteristics of samples collected from the eastern Pyrenees. Notation is the same as for Figure 2.

The highest unblocking temperature component (component C), although not so clearly defined, is identified as stable endpoint directions in the temperature range 450°- 600°C (Figure 4). Above these temperatures the magnetization directions in some samples become erratic, and susceptibility increases, marking the formation of new magnetic minerals which acquire sporadic magnetisations, obscuring the original paleomagnetic directions in the samples; thus component C is not always fully demagnetised (Figure 4). Intensity decay curves for most CCR samples show a gradual decrease in intensity from ~ 250° to 600°C. At lower temperatures, there is occasionally a slight increase in intensity, probably reflecting the removal of the present-day component from the underlying B direction, which is of opposing polarity. In a few samples, however, the presence of a mineral with very low unblocking temperatures (probably goethite) is indicated by a sharp decrease in intensity between room temperature and 150°C. Occasionally, there is also an inflection in the decay curve between 300° and 350°C, indicating a fairly small contribution by an iron sulphide mineral (pyrrhotite) to the NRM. Thermochemical alteration of this mineral at higher temperatures is probably responsible for the spurious magnetisations acquired during the experimental procedure. The gradual decay up to 600°C, by which time most samples have been reduced to < 5% of their original intensity, suggests the predominance of magnetite as the carrier of the high-temperature component. In some cases, up to 10% of the original NRM remains after heating to 600°C, suggesting the contribution of a higher unblocking temperature mineral (haematite) to the NRM.

Component C identified in these samples has declination values ranging from south to west, with scattered positive inclination values in situ, and can be identified in a total of 35 samples from six sites (Figure 6 and Table 2). After bedding correction, the sample directions and site mean directions do not have a true Fisherian distribution but are slightly streaked within the SW quadrant in stereographic projection (Figure 6). This may be related to the rather low intensity of magnetisation of these samples at higher temperatures, making it, in some cases, difficult to clearly identify and isolate the high-temperature C magnetisation. This is particularly the case for some samples from sites CCR5 and CCR6, as reflected in the larger α_{95} values for these two site mean directions. In order to test whether component C has been accurately isolated from the secondary B magnetisation, great circle analysis was carried out. The resulting great circle intersection is not

Table 1. Site Mean Directions for Component B

Site	*N/N**	In situ			Bedding Corrected		
		Dec/Inc	*k*	α_{95}°	Dec/Inc	*k*	α_{95}°
Catalan Coastal Ranges							
SCO1	10/10	181/-03	59.9	6.3	180/+24	29.3	9.1
SCO2	7/7	189/-07	55.5	8.2	190/+22	55.5	8.2
SCO3	7/6	178/+07	47.6	9.8	179/-01	47.6	9.8
SCO4	8/8	182/+12	109.4	5.8	186/+18	109.4	5.8
SCO5	7/5	193/-20	89.8	8.1	241/-44	89.8	8.1
SCO6	7/7	192/-07	102.7	5.5	205/-15	102.7	5.5
SCO7	9/8	192/-04	76.5	6.4	189/+08	76.5	6.4
Mean of seven sites		187/-03	43.1	9.0	195/+02	7.1	24.4

N, number of samples measured; *N**, number of samples used in calculation of site mean; Dec/Inc, declination/inclination (in degrees); *k*, precision parameter (after *Fisher* [1953]); α_{95}, semiangle of the cone of 95% confidence.

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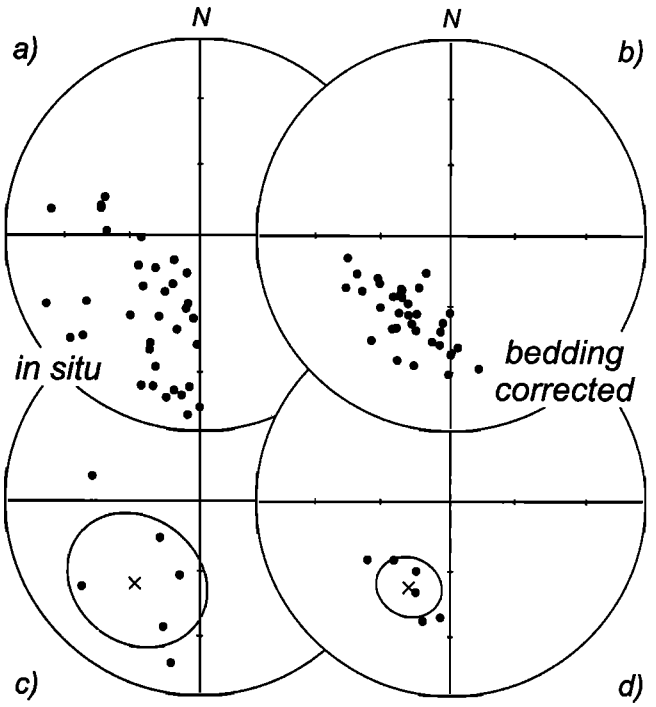


Figure 6. Stereographic projection of the sample directions and site mean directions obtained from the eastern Pyrenees. Notation is the same as for Figure 3.

significantly different from the direction obtained for component C based on linear elements. This provides strong support for our interpretation that component C has been successfully isolated from the overlying secondary B component.

4. Interpretation and Comparison of Results

In the earliest Devonian samples from the Pyrenees the C direction of magnetisation can be identified in a total of 36 samples from seven sites. It yields an in situ mean direction of 307°/58°, k=44.2 and α_{95}=9.2, and 240°/51°, k=86.5 and α_{95}=6.5 after bedding correction (Figure 6). Although there is a clear improvement in the statistical grouping of site mean directions, this is not significant using the fold test of *McElhinny* [1964], with maximum k ratios of 2.06 (N=7) at 90% unfolding. It should be noted, however, that within two sites, there is a slight variation of structural attitude, and in both cases the fold test is positive within site. Also, when the fold test is applied at the sample, rather than site level, it is positive with 99% confidence at 93% unfolding [*McElhinny*, 1964]. An alternative interpretation is that the in situ overall mean direction of 307°/58° is a more recent overprint direction of Cretaceous or younger age. Whilst this is possible, it is not the favored interpretation, given the improvement in statistical parameters after bedding correction both within and between sites.

The southerly and shallow B direction of magnetisation can be identified in seven sites from the CCR and yields an overall site mean direction (7 sites) of 187°/-03°, k=43.1 and α_{95}=9.0° in situ, and 195°/02°, k=7.1, α_{95}=24.4° after bedding correction (Figure 5 and Table 1). It fails the fold test of *McFadden* [1990], demonstrating that acquisition of magnetisation postdates deformation and is of secondary origin. Taking the B direction of magnetisation at face value, it is in good agreement with the late Carboniferous direction expected from the apparent polar wander (APW) path for stable Europe. However, it would reasonably be presumed that in post-Paleozoic times, this region was part of the Iberian plate and thus should reflect counterclockwise rotations of some 35° related to opening of the Bay of Biscay in the late Mesozoic. One possible interpretation, therefore, is that these data indicate that the Catalan terrane was not a stable part of Iberia in post-Variscan times and has undergone a relative, post-Mesozoic clockwise rotation [see e.g., *Parés et al.*, 1988]. Alternatively, the secondary directions identified in this study are similar to Lower Triassic results obtained from central Spain [*Parés and Dinares-Turell*, 1994; *Van der Voo*, 1967; *Osete et al.*, 1997], although as a whole, most Permo-Triassic paleomagnetic data from Iberia show a clear counterclockwise rotation. Such rotations are not apparent in the overprint direction identified in this study. However, whether this is due to thrust sheet rotation, block rotations in relation to Alpine deformation, or simply that this region was not part of the Iberian plate remains equivocal. In summary, before an age and the tectonic implications of the B direction can be described with any confidence, more reliable paleomagnetic data are required from younger sequences in this region, as well as data from the Iberian Peninsula as a whole in order to remove ambiguities within the current paleomagnetic data set for this region in Permo-Triassic times.

Table 2. Site Mean Directions for Component C

Site	*N/N**	In situ			Bedding Corrected		
		Dec/Inc	k	α_{95}°	Dec/Inc	k	α_{95}°
		Catalan Coastal Ranges					
SCO1	10/6	190/+15	79.7	7.6	201/+49	79.7	7.6
SCO2	7/7	196/+32	42.1	9.4	204/+56	42.1	9.4
SCO3	7/6	227/+67	41.4	9.5	235/+46	41.4	9.5
SCO4	8/7	195/+57	52.1	8.4	224/+55	52.1	8.4
SCO5	7/5	283/+42	44.6	11.6	185/+39	44.6	11.6
SCO6	7/5	234/+25	27.6	14.8	193/+36	27.6	14.8
Mean of six sites		218/+44	6.2	29.1	208/+48	28.4	12.8
		Eastern Pyrenees					
BAE2	5/5	287/+54	274.5	4.6	242/+47	274.5	4.6
BAE3	7/7	294/+53	32.5	10.7	243/+49	32.7	10.7
BAE4	5/5	292/+56	490.4	3.5	241/+44	490.4	3.5
BAE6	5/4	318/+57	57.8	12.2	235/+63	209.8	6.4
BAE7	5/5	308/+56	202.4	5.4	249/+52	202.4	5.4
BAE8	7/7	310/+66	145.1	5.0	238/+41	145.1	5.0
BAE9	7/6	348/+55	53.0	9.3	230/+59	142.8	5.6
Mean of seven sites		307/+58	44.2	9.2	240/+51	86.5	6.5
Mean of all 13 sites		262/+61	5.5	19.5	224/+51	24.6	8.5

N, number of samples measured: *N**, number of samples used in calculation of site mean: Dec/Inc, declination/inclination (in degrees): k, precision parameter (after *Fisher* 1953): α_{95}, semi-angle of the cone of 95% confidence.

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The C magnetisation direction from the CCR is identified with single polarity, in 34 samples from six sites, yielding an overall site mean direction of 218°/44°, k=6.2 and α_{95}=29.1° in situ, and 208°/48°, k=28.4 and α_{95}=12.8° after bedding correction (Figure 6 and Table 2). The improvement in statistical parameters is significant at the 95% level of confidence using the test of both *McFadden* [1990] and *McElhinny* [1964]. This is taken as a good indication that the high-temperature C component of magnetisation is prefolding in origin. The apparent streaking of the data obtained from these samples (Figure 5) could be taken as indication that in some samples the B component has not been completely isolated from the high-temperature component. It is also possible that there has been small-scale intraformational rotation in this thrusted sequence. This could explain the apparent smearing of the bedding corrected CCR5 and CCR6 data along a small circle. These two sites were collected from the upper part of the sequence. However, taking into account the great circle analysis (see above), the significance of the fold test at site level, and that the C component is in most cases identified at higher temperatures as a stable endpoint direction after the removal at intermediate temperatures of component B, we feel confident that the overall mean direction obtained for C does approximate to the primary direction of magnetisation.

The two sampling regions in this study are situated in rather different tectonic settings, and differential rotations of the sampled units may be expected. While we are aware that, strictly speaking, it may not be correct to combine the data from the two regions which may have undergone different deformational histories, given that both sets of data yield positive fold tests, it was decided to combine the data at the site level in order to carry out a regional fold test. This results in an overall in situ mean of 262°/61°, k=5.5 and α_{95}=19.5° for component C, and 224°/51°, k=24.6 and α_{95}=8.5° (13 sites) after bedding correction (Figure 7). The improvement in statistical parameters is positive at the 99% level of confidence using the fold tests of both *McElhinny* [1964] and *McFadden* [1990] and at 90% unfolding. This is quite compelling evidence that component C from both regions is the primary direction of magnetisation. Component C, therefore, translates into paleolatitudes of ~ 30°S for the Catalan Coastal Ranges and eastern Pyrenees in Siluro-Devonian times.

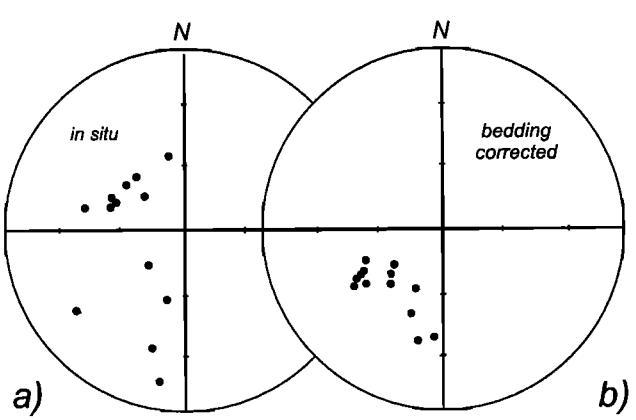


Figure 7. Combined site mean directions from the Catalan Coastal Ranges and the eastern Pyrenees. Notation is the same as for Figure 3.

The declination values indicate a small amount of differential rotation between the two regions. This may be related to differential rotation of the two regions during either Variscan or Alpine deformation. No further interpretation of the declination values can be attributed to the results, however, due to problems already mentioned with regards to the post-Paleozoic tectonic development in this part of the Iberian Peninsular. First, it is not clear if this part of the eastern Pyrenees was part of the Iberian plate in Mesozoic times or if it was coherent with stable Europe. Furthermore, rotation of Catalunya in post-Eocene times has been suggested. The lack of well-defined overprint directions and the lack of control on the tectonic coherence of the regions does not allow a valid interpretation of the declination values obtained.

5. Conclusions

A high-temperature paleomagnetic direction is identified in a total of 13 sites (75 samples) from Siluro-Devonian boundary sediments in the Catalan Coastal Ranges and Axial Zone of the eastern Pyrenees of NE Spain. Combining the site mean directions from both regions yields a direction of 224°/51°, k=24.5 and α_{95}=8.5° after bedding correction. The directions pass the fold test with 99% confidence, indicating the primary nature of this component of magnetisation. The declination value is difficult to accurately interpret, given the possibility of younger, but as yet not clearly quantified, rotations of the area. The inclination value, however, demonstrates that both regions were at paleolatitudes of some 30°S in latest Silurian to Early Devonian times (Figure 8). This is similar to paleomagnetic results obtained from Late Silurian rocks from the Tepla-Barrandian of the Bohemian Massif [*Tait et al.*, 1994] and from Early Devonian rocks of the Armorican Massif [*Tait, 1999*]. These two massifs formed part of the Armorican Terrane Assemblage (ATA), i.e., the Paleozoic Gondwana-derived terranes which today constitute part of the Variscan fold belt of Europe, south of the Rhenohercynian Zone. These terranes were previously considered to have formed part of a coherent microplate (the Armorican microplate), but from paleomagnetic and geological evidence it is now clear that the concept of a coherent Armorican microplate is no longer valid and the various blocks were separated by minor oceanic basins [*Tait et al.*, 1997; *Franke et al.*, 1995]. Although there are no reliable paleomagnetic data for the Siluro-Devonian of the Iberian Massif, on faunal and lithological grounds it is fairly clear that the Iberian and Armorican Massifs were closely related throughout the Paleozoic [*Robardet and Guiterrez-Marco*, 1990; *Quesada et al.*, 1991]. Thus the different units of the ATA all appear to have had similar northward drift histories during the Paleozoic and in Late Silurian times were all situated between paleolatitudes of 25° and 30°S (Figure 8). The results presented here from northeastern Spain demonstrate that this region was adjacent to the southern margin of the Armorican Terrane Assemblage and have important implications for the geodynamic evolution of this region in Paleozoic times, although the tectonic relationship between this region and the Iberian Massif remains equivocal due to the lack of paleomagnetic data from Spain and Portugal during the Paleozoic.

Although the paleopositions of Baltica, Siberia, and Laurentia are all well constrained for the Paleozoic as a whole

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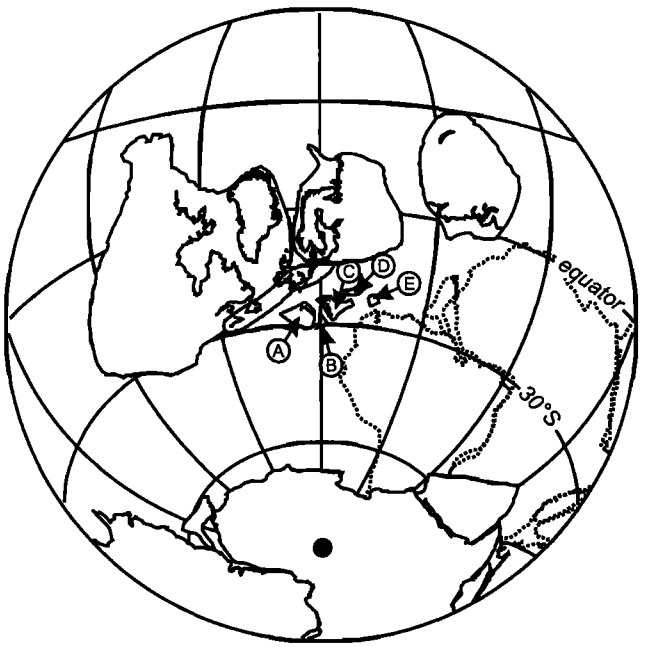


Figure 8. Palaeogeographic reconstruction for Siluro-Devonian boundary times based on data from the present study, *Torsvik et al.* [1992] for Baltica, *Channell et al.* [1992] for Avalonia, *Bachtadse et al.* [2000] for Siberia, *Tait* [1999] for the Armorican Massif, and *Tait et al.* [1994] for the Barrandian Zone of the Bohemian Massif. Two different options for the position of Gondwana are indicated. The dotted outline indicates the palaeoposition according to the more complex APW path based on the paleomagnetic data of *Schmidt et al.* [1990]. The solid outline indicates the preferred position using the more conservative APW path of *Bachtadse and Briden* [1991]. A, Iberian Massif; B, Catalunya and Eastern Pyrenees; C, Armorican Massif and French Massif Central; D, Saxothuringia; E, Barrandia.

(see review papers by *MacNiocall and Smethurst* [1994]; *Torsvik et al.*, [1992, 1996] and *Smethurst et al.*, [1998]), paleogeographic reconstructions for mid-Paleozoic times remain contentious with regards to Gondwana [*Van der Voo* [1993]; *Bachtadse et al.*, 1995]. Basically, there are two different options from the existing paleomagnetic data set. A more complicated APW path for Gondwana can be adopted (path Y after *Morel and Irving* [1978], and subsequently modified by *Schmidt et al.*, [1990]), in which case the northern margin of Gondwana would have been in an equatorial position in Siluro-Devonian boundary times (see dotted outline in Figure 8). The alternative APW path (path X of *Morel and Irving* [1978], more recently redefined by *Bachtadse and Briden* [1991]) involves gradual northward movement of Gondwana throughout the Palaeozoic and places the northern margin of Gondwana at approximately 65°S at the Siluro-Devonian boundary (see solid outline in Figure 8). Whichever of these two models is adopted, it is clear from the new palaeomagnetic data that the ATA, including the Catalan terrane, did not remain adjacent to the north African margin of Gondwana. Alternative paleogeographies based on biogeographic and climatic evidence place the North African margin of Gondwana between ~ 50° and 30°S in the Siluro-Devonian [*Scotese et al.*, 1999, *Scotese and McKerrow;* 1991 and references therein, *McKerrow et al.*, 2000]. None of these biogeographic models are in apparent direct conflict with the new palaeomagnetic data presented here from NE Spain.

Acknowledgments. The authors would like to acknowledge financial support from the Deutsche Forschungsgemeinschaft (Ta193/1). Many thanks to Josep Parés and Oriol Oms for their help and support in the field. Thanks also to the reviewers Bernard Henry and Trond Torsvik, whose comments helped improve the manuscript.

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(Received June 28 1999; revised February 22, 2000;
accepted April 3, 2000.)

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